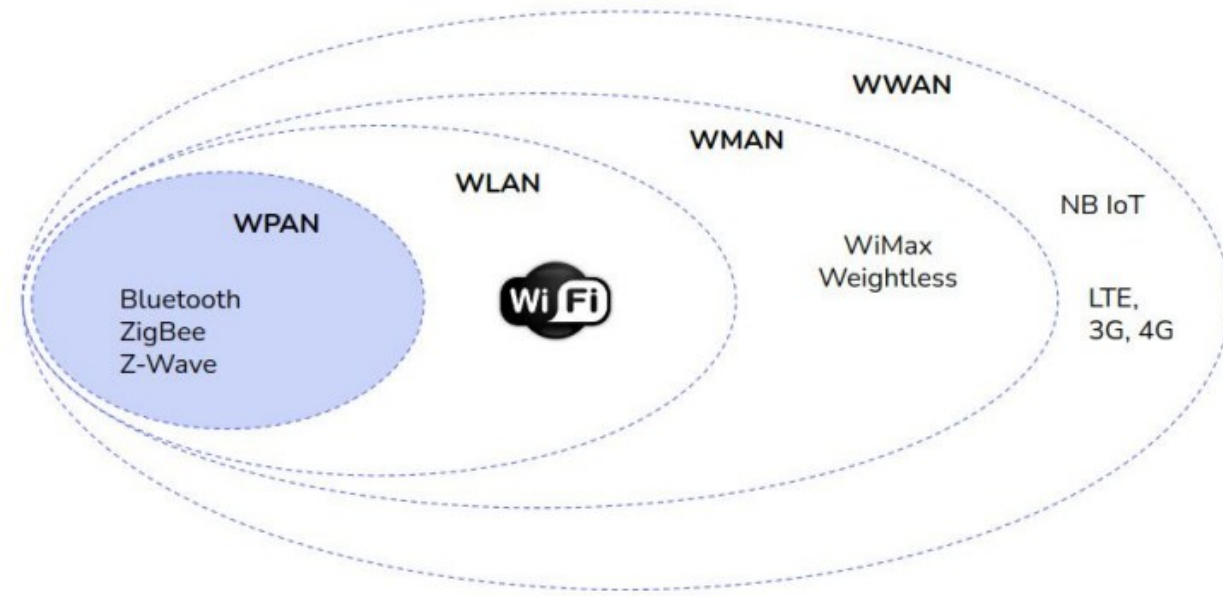


IoT Networking - Protocol landscape

OSI Layer	Role in IoT Networking	Common Protocols / Technologies
Physical Layer	Transmission of raw bits over physical medium	Wired: Ethernet, RS232, RS485, USB Wireless: Bluetooth, BLE, ZigBee, Z-Wave, Wi-Fi, LoRaWAN, GSM (2G/3G/4G/LTE), 5G, NB-IoT
Data Link Layer	Node-to-node data transfer, MAC addressing	Wi-Fi MAC, ZigBee MAC, Bluetooth MAC, Ethernet MAC
Network Layer	Addressing and routing of data packets	IPv4, IPv6
Transport Layer	Reliable or fast delivery of data	TCP (reliable), UDP (fast, lightweight), RTSP (real-time streaming)
Application Layer	Defines how applications exchange data	HTTP, HTTPS, Web Services, MQTT (publish-subscribe), CoAP (lightweight for constrained devices), Modbus, FTP, SMTP, XMPP

IoT Networking - Wireless Landscape



Wireless

- ✓ WPAN - PAN 802.15
- ✓ WLAN - LAN 802.11
- ✓ WMAN - MAN 802.16
- ✓ WWAN - WAN 802.20

- **Zigbee** is an IEEE 802.15.4-to create personal area networks. Zigbee is a low-power, low-data-rate, and close proximity wireless ad hoc network.
- **Z-Wave** is a wireless communications protocol used primarily for residential and commercial building automation.
- **Wimax**, short for Worldwide Interoperability for Microwave Access, is a type of wireless communication technology that enables high-speed internet access over short distances.
- **Weightless** technology delivers wireless connectivity for low power, wide area networks (LPWAN) specifically designed for the Internet of Things.
- **NarrowBand-Internet of Things (NB-IoT)** is a standards-based low power wide area (LPWA) technology developed to enable a wide range of new IoT devices and services.

IoT Networking - Wireless Landscape

Trends & Shifts

Overall, the landscape is shifting away from a one-size-fits-all approach. Instead, we're moving toward **specialized connectivity tailored for specific use cases**, often in hybrid configurations. This ensures IoT systems can meet diverse requirements—whether it's low power, long range, high bandwidth, or ultra-reliable communication.



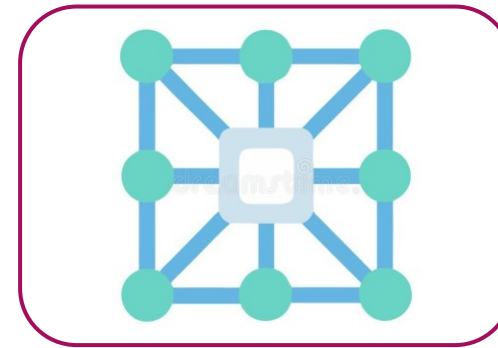
Low-Power Wi-Fi

Features like Target Wake Time in Wi-Fi 6 allow Wi-Fi to enter the low-power space.



Hybrid Networking

Successful IoT deployments rarely use one protocol, opting instead to combine technologies for flexibility.



Mesh Architectures

Zigbee, Z-Wave, and Thread provide distributed, self-healing connections for improved reliability.

Networking Protocol Design

What is a Protocol ? -

- Protocol– means **Medium** and **rules to connect, communicate and exchange data** .
- Designed for Device-to-device communication - Influenced by application design, use case; devices setup and numerous other factors.
- Need to identify the device which sends or receives data.
- Common “Spoken language” of devices - Set of rules & regulations
- Command and Data transfer - Efficient, reliable and fault-tolerant
- Protocol can be at various Levels - Hardware, Software, Application
- Examples: MQTT, COAP, HTTP, AMQP, LwM2M

Each protocol has 3 basic Elements

- Syntax - -Structure of data
- Semantics - Meaning of the data
- Timing - Speed, Ordering, Synchronization

Popular Protocols

1. MQTT
2. AMQP
3. Bluetooth
4. ZigBee
5. WiFi
6. Cellular



- COAP – Constrained Application Protocol
- LwM2M – Light Weight Machine to Machine Protocol

Networking Protocol Design Considerations



Addressing the systems

- IP Address OR MAC Address OR URLs



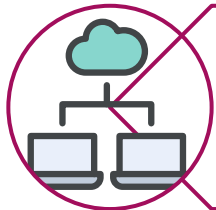
Choice of transport

- Peer to Peer OR over the internet



Error Handling

- Retries OR Data errors



Payload

- XML, JSON OR Files



Message pattern

- Pub-Sub (Data Push) – MQTT
- Request-Response – HTTP o RPC



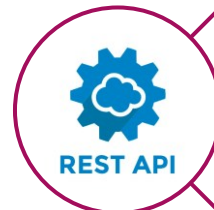
Quality of Service (QoS)

- Messaging and delivery reliability
- Retry mechanisms ; Storage till delivery



Security

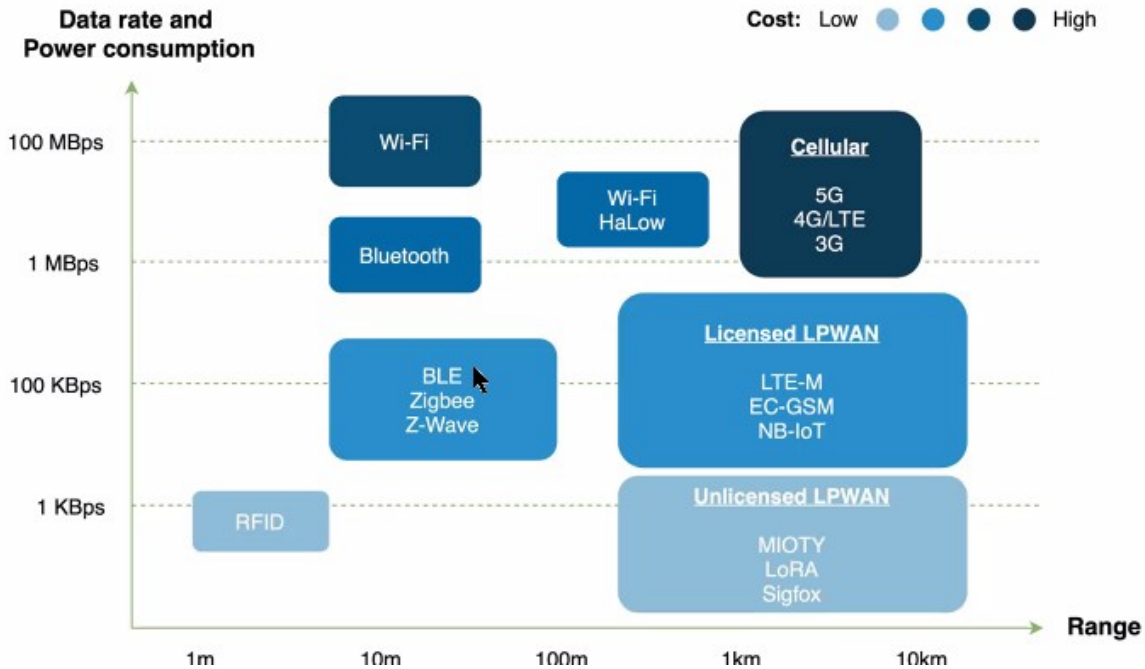
- Transport level & Application level



RESTful or RESTless

- XML, JSON OR Files

Networking Protocol Design Considerations



Technology	Frequency	Data Rate	Range	Power Usage	Cost
Bluetooth LE	2.4 GHz ISM	Up to 2 Mbps	1–100 m	Very low (<15 mA TX)	Low (\$1–\$2)
Zigbee (802.15.4)	2.4 GHz / Sub-GHz	20–250 kbps	10–100 m (mesh extends)	Low (battery lasts years)	Low (~\$2)
Z-Wave	Sub-GHz (868/915 MHz)	~100 kbps	30–100 m	Low	Low (~\$2–\$3)
Wi-Fi (802.11)	2.4 / 5 / 6 GHz	7 Mbps–9.6 Gbps	~25 m indoors	Moderate (hundreds of mA)	Moderate to high
WirelessHART	2.4 GHz	~250 kbps	30–200 m (mesh)	Low	Moderate
LoRaWAN	Sub-GHz (868/915 MHz)	0.3–50 kbps	5–10 km (rural >10 km)	Ultra-low (sleep most time)	Low
Sigfox	Sub-GHz (868/915 MHz)	~100 bps	10–50 km	Ultra-low (years on battery)	Low
Weightless	Sub-GHz	~100 kbps	Up to 5 km	Ultra-low	Low
NB-IoT	Licensed LTE bands	~20–200 kbps	1–10 km	Very low (battery up to 10 yrs)	Low (<\$5 module)
LTE-M (Cat-M1)	LTE bands (~1.4 MHz)	Up to 1 Mbps	10–40 km	Low to moderate (PSM/eDRX)	Moderate
2G/3G	Licensed cellular bands	~100 kbps–2 Mbps	Wide area (km-level)	Moderate (higher than LPWAN)	Moderate
LTE (4G)	Licensed cellular bands	Up to 100 Mbps	Wide area	High (compared to LPWAN)	Higher